

Linear Regression

egg 2026, experimental semantics, day 4

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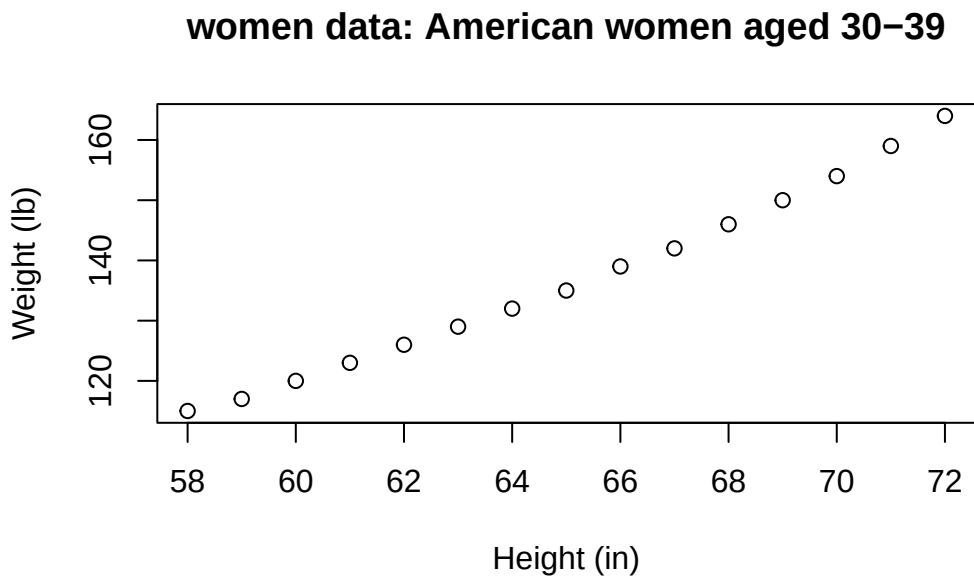
1 History

- Laplace, Gauss, ...
- [least squares method](#)
- Gauss and astronomy: [Ceres](#)

2 Non-linguistic example

```
require(graphics)

plot(women, xlab = "Height (in)", ylab = "Weight (lb)",
      main = "women data: American women aged 30-39")
```



```
data("women")

head(women)
```

	height	weight
1	58	115
2	59	117
3	60	120
4	61	123
5	62	126
6	63	129

```
help("women")
```

Average Heights and Weights for American Women Description This data set gives the average heights and weights for American women aged 30-39.

Usage women Format A data frame with 15 observations on 2 variables.

[,1] height numeric Height (in) [,2] weight numeric Weight (lbs) Details The data set appears to have been taken from the American Society of Actuaries Build and Blood Pressure Study for some (unknown to us) earlier year.

The World Almanac notes: “The figures represent weights in ordinary indoor clothing and shoes, and heights with shoes”.

Source The World Almanac and Book of Facts, 1975.

References McNeil, D. R. (1977) Interactive Data Analysis. Wiley.

Examples require(graphics) plot(women, xlab = “Height (in)”, ylab = “Weight (lb)”, main = “women data: American women aged 30-39”) [Package datasets version 4.0.5 Index]

- and now a model that captures an intuitively clear relationship between height and weight
- height is the *explanatory* variable
- weight is the *dependent* variable
- alternative terminology: explanatory vs. independent variables
- model:

```
lm <- lm(women$weight ~ women$height)
```

```
lm
```

Call:

```
lm(formula = women$weight ~ women$height)
```

Coefficients:

(Intercept)	women\$height
-87.52	3.45

- y-intercept (-87.52)
- a woman with zero height would weigh -87.52 pounds
- regression coefficient (3.45)
- a numerically expressed relationship between the explanatory and dependent variables
- the increase in the dependent variable when the explanatory variable increases by 1 unit

```
summary(lm)
```

```

Call:
lm(formula = women$weight ~ women$height)

Residuals:
    Min       1Q   Median       3Q      Max
-1.7333 -1.1333 -0.3833  0.7417  3.1167

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -87.51667     5.93694   -14.74 1.71e-09 ***
women$height   3.45000     0.09114    37.85 1.09e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.525 on 13 degrees of freedom
Multiple R-squared:  0.991, Adjusted R-squared:  0.9903
F-statistic: 1433 on 1 and 13 DF,  p-value: 1.091e-14

```

- each coefficient has:
 1. sign: positive or negative
 2. magnitude (effect size)
 3. significance: probability of the null hypothesis given the sample
- with a sufficiently large number of observations, thanks to the [central limit theorem](#), we should not end up too far away (standard error) from the true population mean
- R^2 measures model fit
- in this very limited sample, the model explains 99%, which is unrealistic for a typical sample
- of course, many more factors matter: age, sex, ...
- linear regression coefficients also determine the fitted line
- [linear regression](#)
- and also residuals for each observation

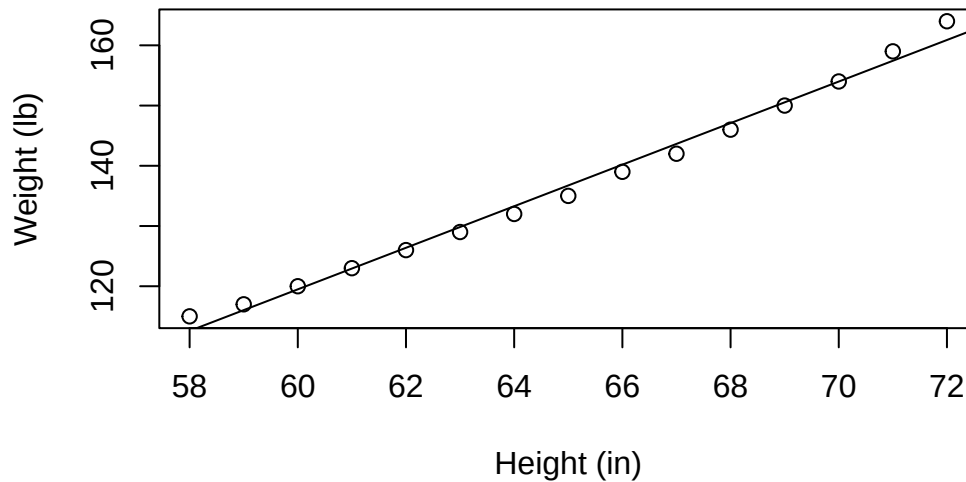
```

plot(women, xlab = "Height (in)", ylab = "Weight (lb)",
     main = "women data: American women aged 30-39")

abline(lm(women$weight ~ women$height))

```

women data: American women aged 30–39



3 Multiple regression analysis

- a slightly more realistic example
- data from this [link](#)

```
heart.data <- read.csv("data/heart.data.csv", header = TRUE)

heart.data$X <- NULL

heart.data$biking <- as.numeric(heart.data$biking)
heart.data$smoking <- as.numeric(heart.data$smoking)
heart.data$heart.disease <- as.numeric(heart.data$heart.disease)

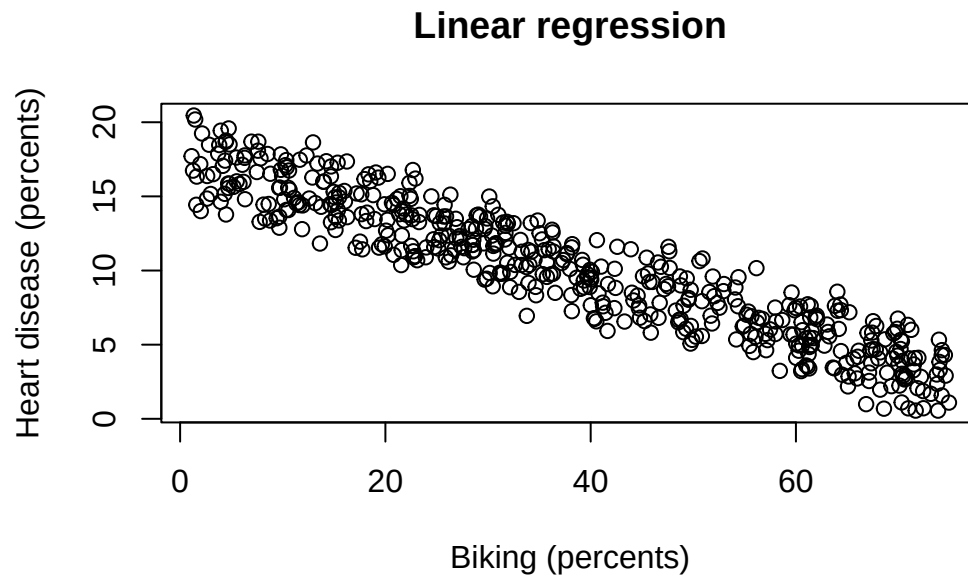
head(heart.data)
```

	biking	smoking	heart.disease
1	30.801246	10.896608	11.769423
2	65.129215	2.219563	2.854081
3	1.959665	17.588331	17.177803
4	44.800196	2.802559	6.816647
5	69.428454	15.974505	4.062224
6	54.403626	29.333176	9.550046

- city-level statistics and percentages of cyclists, smokers, and people with heart disease

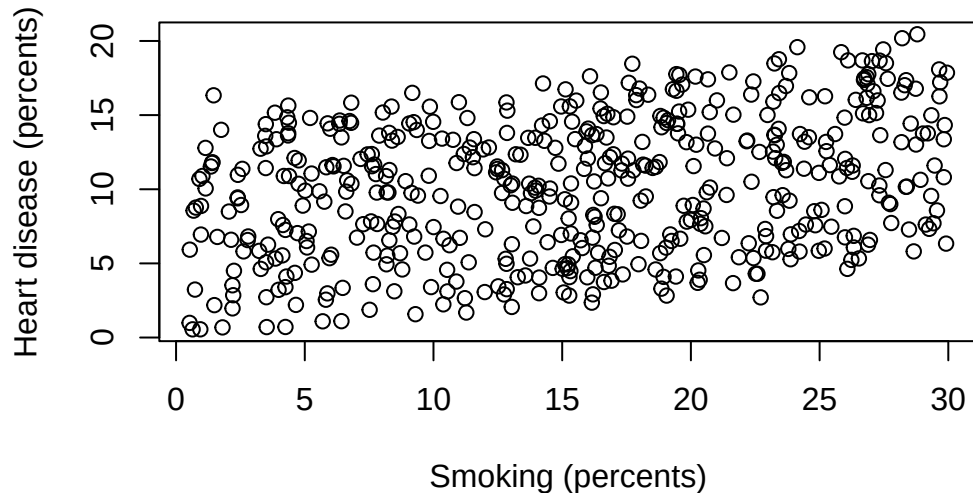
- two explanatory variables
- one dependent variable
- multiple linear regression
- correlation plots:

```
plot(heart.data$heart.disease ~ heart.data$biking, xlab = "Biking (percents)", ylab = "Heart disease (percents)",  
     main = "Linear regression")
```



```
plot(heart.data$heart.disease ~ heart.data$smoking, xlab = "Smoking (percents)", ylab = "Heart disease (percents)",  
     main = "Linear regression")
```

Linear regression



- a similarly constructed model, but with two explanatory variables

```
heart.disease.lm <- lm(heart.disease ~ biking + smoking, data = heart.data)
heart.disease.lm
```

Call:

```
lm(formula = heart.disease ~ biking + smoking, data = heart.data)
```

Coefficients:

(Intercept)	biking	smoking
14.9847	-0.2001	0.1783

- more detailed output:

```
summary(heart.disease.lm)
```

Call:

```
lm(formula = heart.disease ~ biking + smoking, data = heart.data)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.1789	-0.4463	0.0362	0.4422	1.9331

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.984658	0.080137	186.99	<2e-16 ***
biking	-0.200133	0.001366	-146.53	<2e-16 ***
smoking	0.178334	0.003539	50.39	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.654 on 495 degrees of freedom

Multiple R-squared: 0.9796, Adjusted R-squared: 0.9795

F-statistic: 1.19e+04 on 2 and 495 DF, p-value: < 2.2e-16

```
# library(ggplot2)
# abline(lm(heart.disease ~ biking, data = heart.data))
```

- simpler model:

```
heart.disease.lm2 <- lm(heart.disease ~ biking, data = heart.data)

heart.disease.lm2
```

Call:

```
lm(formula = heart.disease ~ biking, data = heart.data)
```

Coefficients:

(Intercept)	biking
17.6979	-0.1991

```
summary(heart.disease.lm2)
```

Call:

```
lm(formula = heart.disease ~ biking, data = heart.data)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.028	-1.206	-0.004	1.151	3.643

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	17.697884	0.146780	120.57	<2e-16 ***
biking	-0.199091	0.003378	-58.94	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

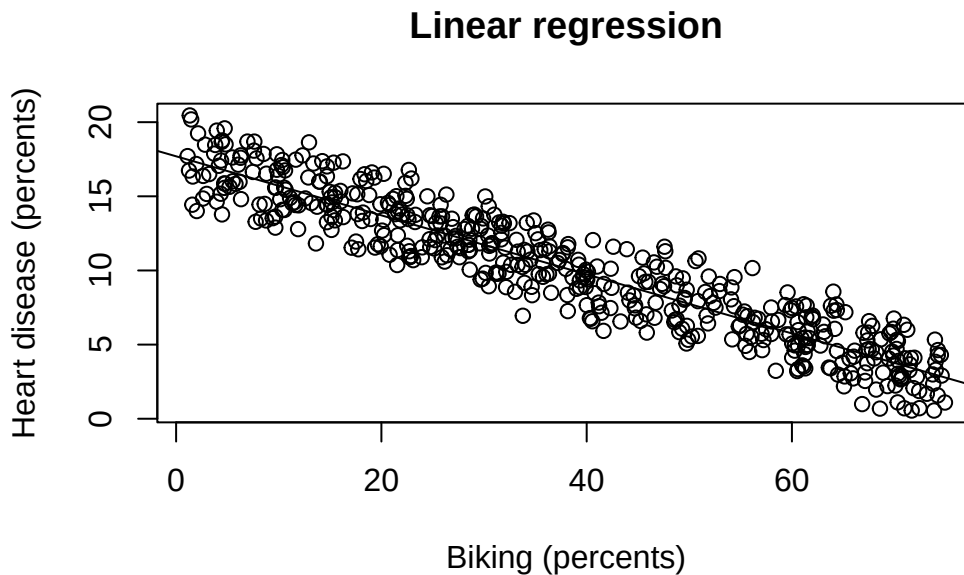
Residual standard error: 1.618 on 496 degrees of freedom

Multiple R-squared: 0.8751, Adjusted R-squared: 0.8748

F-statistic: 3474 on 1 and 496 DF, p-value: < 2.2e-16

- worse R^2
- plot of simpler model 3:

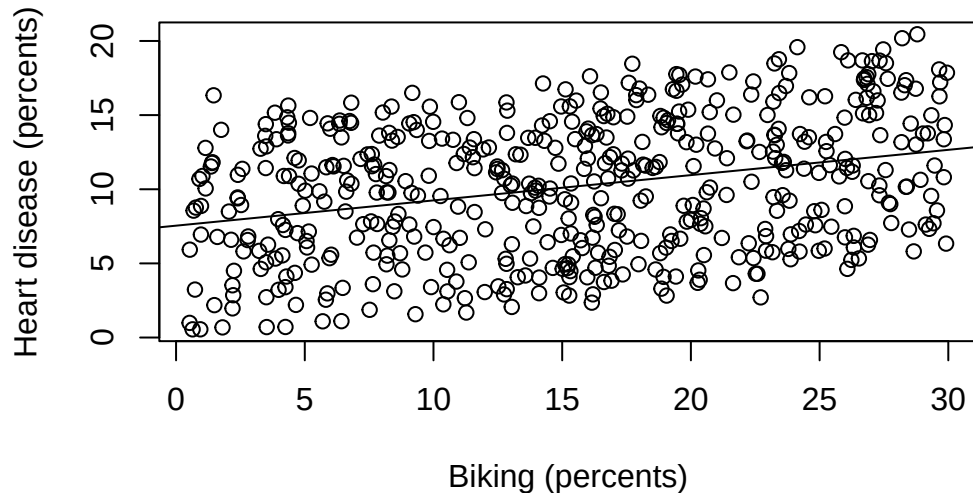
```
plot(heart.data$heart.disease ~ heart.data$biking, xlab = "Biking (percents)", ylab = "Heart
      disease (percents)", main = "Linear regression")
abline(lm(heart.data$heart.disease ~ heart.data$biking))
```



- plot of simpler model 2:

```
plot(heart.data$heart.disease ~ heart.data$smoking, xlab = "Biking (percents)", ylab = "Heart
      disease (percents)", main = "Linear regression")
abline(lm(heart.data$heart.disease ~ heart.data$smoking))
```

Linear regression



```
heart.disease.lm3 <- lm(heart.disease ~ smoking, data = heart.data)
heart.disease.lm3
```

Call:

```
lm(formula = heart.disease ~ smoking, data = heart.data)
```

Coefficients:

(Intercept)	smoking
7.5431	0.1705

```
summary(heart.disease.lm3)
```

Call:

```
lm(formula = heart.disease ~ smoking, data = heart.data)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.7065	-3.7069	0.5007	3.6597	8.5434

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.54311	0.41251	18.286	< 2e-16 ***

```
smoking      0.17048    0.02355    7.239 1.73e-12 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 4.352 on 496 degrees of freedom

Multiple R-squared: 0.09556, Adjusted R-squared: 0.09374

F-statistic: 52.41 on 1 and 496 DF, p-value: 1.729e-12

- model comparison using ANOVA

```
anova(heart.disease.lm2, heart.disease.lm)
```

Analysis of Variance Table

Model 1: heart.disease ~ biking

Model 2: heart.disease ~ biking + smoking

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	496	1297.74				
2	495	211.74	1	1086	2538.8	< 2.2e-16 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- the more complex (Df) model is much better
- comparison of the two simpler models

```
anova(heart.disease.lm3, heart.disease.lm2)
```

Analysis of Variance Table

Model 1: heart.disease ~ smoking

Model 2: heart.disease ~ biking

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	496	9395.6				
2	496	1297.7	0	8097.8		

- the model using biking as the explanatory variable is much better
- back to the complex model

```
summary(heart.disease.lm)
```

```
Call:
lm(formula = heart.disease ~ biking + smoking, data = heart.data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2.1789	-0.4463	0.0362	0.4422	1.9331

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.984658	0.080137	186.99	<2e-16 ***
biking	-0.200133	0.001366	-146.53	<2e-16 ***
smoking	0.178334	0.003539	50.39	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.654 on 495 degrees of freedom

Multiple R-squared: 0.9796, Adjusted R-squared: 0.9795

F-statistic: 1.19e+04 on 2 and 495 DF, p-value: < 2.2e-16

```
# library(ggplot2)
# abline(lm(heart.disease ~ biking, data = heart.data))
```

- individual coefficients are computed while holding the other factors fixed
- e.g., how much biking improves heart disease outcomes when smoking is held at the same value

4 Linguistic example

- a linear model from Hanka's master's thesis

```
body {
  counter-reset: example-counter 0;
}

.ex::before {
  counter-increment: example-counter;
  content: "(" counter(example-counter) ") ";
}

.ab::before {
```

```

    counter-increment: ab-counter;
    content: counter(ab-counter, lower-latin) ".";
}

.example-number {
    width: 2em;
}

.ab-counter {
    width: 2em;
}

.judgement {
    text-align: right;
    width: .5em;
    padding-right: 0.5em;
}

.smallcaps {
    font-variant: small-caps;
}

.example-container {
    counter-reset: ab-counter 0;
    line-height: 1.1;
}

.individual-example {
    display: flex;
    flex-direction: row;
    align-items: flex-start;
}

.gloss-example-container {
    counter-reset: ab-counter 0;
}

.gloss-individual-example {
    display: flex;
    flex-direction: row;
    align-items: flex-start;
    margin-bottom: -1em;
}

```

```

}

.gloss-example {
  overflow: auto;
  width: 90%;
  margin-left: -2.5em;
}

.gloss-individual-word {
  margin-top: 0;
}

ol.sentence {
  display: inline-block;
  list-style-type: none;
  padding: 0;
  width: 90%;
}

.target-word,
.target-gloss {
  list-style-type: none;
}

ol.word {
  margin-bottom: .5em;
  float: left;
  display: inline-block;
}

p.translation {
  width: 90%;
  float: left;
  display: inline-block;
  margin-top: -1em;
  margin-left: 2.5em;
}

```

4.1 Comparatively and superlatively modified numerals (theories, contrasts)

4.1.1 Background

- the accepted contrasts between comparative modifiers and superlative modifiers (see Buring 2008, Geurts 2007, Nouwen 2015, Mayr 2013, Schwarz 2016, among others):
 1. comparative modifiers do not but superlative modifiers do give rise to obligatory ignorance implicatures (*?I have at most three bikes* vs. *I have less than four bikes*);
 2. comparative modifiers can scope over or under existential modals (EM) but superlative modifiers have to outscope them (*The cup of Darjeeling tea can contain less than 50mg of caffeine*: $\Diamond > 50$ ok vs. *The cup of Darjeeling tea can contain at most 50mg of caffeine*: $*\Diamond > 50$)
- a *no more than Num* construction like in English (1a), following Nouwen 2008, is then claimed to:
 1. allow both scopes with respect to existential modals (Nouwen 2008): (1a) – $\Diamond > 20/20 > \Diamond$;
 2. have scalar bounding inference (50 for (1b));
- *no more than Num* construction is then claimed to be:
 1. a subtype of differential quantifiers in the class of comparative modifiers (as *slightly less* in (1c));
 2. supported by the comparative morphology of *no more than Num*;

4.1.2 Our claims

- we bring **new experimental evidence** (from Czech) against such claims, showing that (unlike in English):
 1. *no more than Num* prefers to be interpreted with wider scope than existential modals;
 2. *no more than Num* can behave unlike other differential comparative modifiers;
 3. our experiment shows there are two kinds of differential quantifiers: comparative ((1c) and English (1a)) and superlative (Czech *no more than Num* as in (2a)/(2c));

Comparative and superlative differentials: experimental evidence from Czech - Sinn und Bedeutung 26, Univer...
comparative_superlative_diffs.pdf 25 z 48

Example item

Context: Alex is reading the following sentence on a chocolate bar packaging:

(17) *Toto balení může obsahovat*
this packaging can contain

a. *méně než* FEWER
fewer than

b. *nanejvýš* AT-MOST
at-most

25

Comparative and superlative differentials: experimental evidence from Czech - Sinn und Bedeutung 26, Univer...
comparative_superlative_diffs.pdf 26 z 48

Example item

Context: Alex is reading the following sentence on a chocolate bar packaging:

(18) *Toto balení může obsahovat*
this packaging can contain

a. *ne víc než* NO MORE
no more than

b. *trochu méně než 60 gramů cukru* SLIGHTLY LESS
slightly less than 60 grams of-sugar

Alex says: 'So, in this chocolate bar there can be sometimes even 65 grams of sugar.'

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4.2 Experiment

- two experiments to target two research questions:
 1. does Czech *no more than* behave more like a comparative modifier or a superlative modifier (in the modal environment)?
 2. does Czech *no more* behave like other differential quantifiers?

4.2.1 Design of both experiments

- Czech native speakers
- Likert scale 1-5
- the appropriateness of one of the conditions in a context
- further, only exp 2: it included all the conditions of exp 1
- truth-value judgment task where a context described a situation strongly preferring the wide scope of the existential modal over the degree quantifiers
- 16 items and 16 fillers

- 98 subjects (all of them passed fillers uncontroversial TVJT)
- implemented on L-Rex

The experiment followed an observation from Geurts 2007 and Blok 2019, among others:

- comparative modifiers allow both wide and narrow scope with respect to an existential modal reading but superlative modifiers have to out-scope the existential modals (split-scope)
- four conditions:
 1. comparative modifiers: FEWER, (2a)
 2. superlative modifiers: AT-MOST, (2b)
 3. *no more* modifier: NO-MORE, (2c)
 4. differential *slightly less* modifier, (2d) – SLIGHTLY-LESS

4.2.2 Assumptions

- the conditions FEWER and AT-MOST tested the acceptability of modified numerals without differential
- the conditions SLIGHTLY-LESS and NO-MORE tested the presence of a differential (zero degree differential in the case of NO-MORE)
- all conditions tested possible narrow scope with respect to the existential modal
- Nouwen 2008 predicts NO-MORE to be comparative modifiers like SLIGHTLY-LESS
- the design was a 2x2 factorial: comparative modifiers or superlative modifiers (CLASSA, CLASSB) x absence/presence of a differential (DIFFYES, DIFFNO)
- an example item from the experiment is in (2)

(1)

- a. Cody's paper is allowed to have no more than 20 pages.
- b. No fewer than fifty people showed up.
- c. John is slightly less tall than Mary.

4.2.3 Predictions

- in the existential modal environment manipulated in a way strongly preferring wide scope of the modal
- $N(o)M(ore\ than)C(onstruction)$
- $C(omparative)M(odifier)$
- $S(uperlative)M(odifier)$

	FEWER	AT-MOST	NO-MORE	SLIGHTLY-LESS
PredictionsNMC	✓		✓	✓
as CM				
NMC	✓			✓
as SM				

4.2.4 Results

- the data was analyzed in a mixed-effects linear model with subject and item intercept+slope random effects (R package LMERTEST)
 - the dependent variable: the subject's response
 - several models, the best: conditions as independent variables, plus their interaction
 - the less fitting models included models with main effects only and models where *no more* was treated as comparative modifiers
1. we found a negative main effect of CLASSB (superlative modifiers) (t-value: -11.004, $p < 0.001$) and a positive effect of the absence of a differential (t-value: 3.946, $p < 0.001$)
 2. the model also reports a negative interaction of CLASSB (superlative modifiers) by DIFFNO (t-value: -3.129, $p = 0.002$)
 3. Tukey's pairwise comparison of the conditions reveals that only AT-MOST and NO-MORE were statistically non-significantly different (t-value: -0.478, $p = 0.964$)
- all other pairs of conditions differed significantly
 - the boxplot representing means and standard errors is in Fig. 2.1
 - the experiment thus confirms:
1. the scope behavior of Czech *no more* construction follows the pattern of superlative modifiers, not comparative modifiers, since subjects accepted NO-MORE to the same extent as AT-LEAST
 2. the significant difference between NO-MORE and SLIGHTY-LESS can be explained by classifying *no more* as a superlative-modifier differential quantifier and *slightly less* as a comparative modifier

The surprising result of this experiment is the overall low acceptability of all conditions:

- even the most default comparative modifier without a differential (condition FEWER) had $\mu = 2.51$ (SD: 1.61, SE: 0.04)
- we hypothesize that this results from the priming effect of the most frequent everyday contexts, which strongly prefer the $max_d > \diamond$ reading, the opposite of the contexts described in the experiment

```
knitr::include_graphics("assets/plot_zoom.png")
```

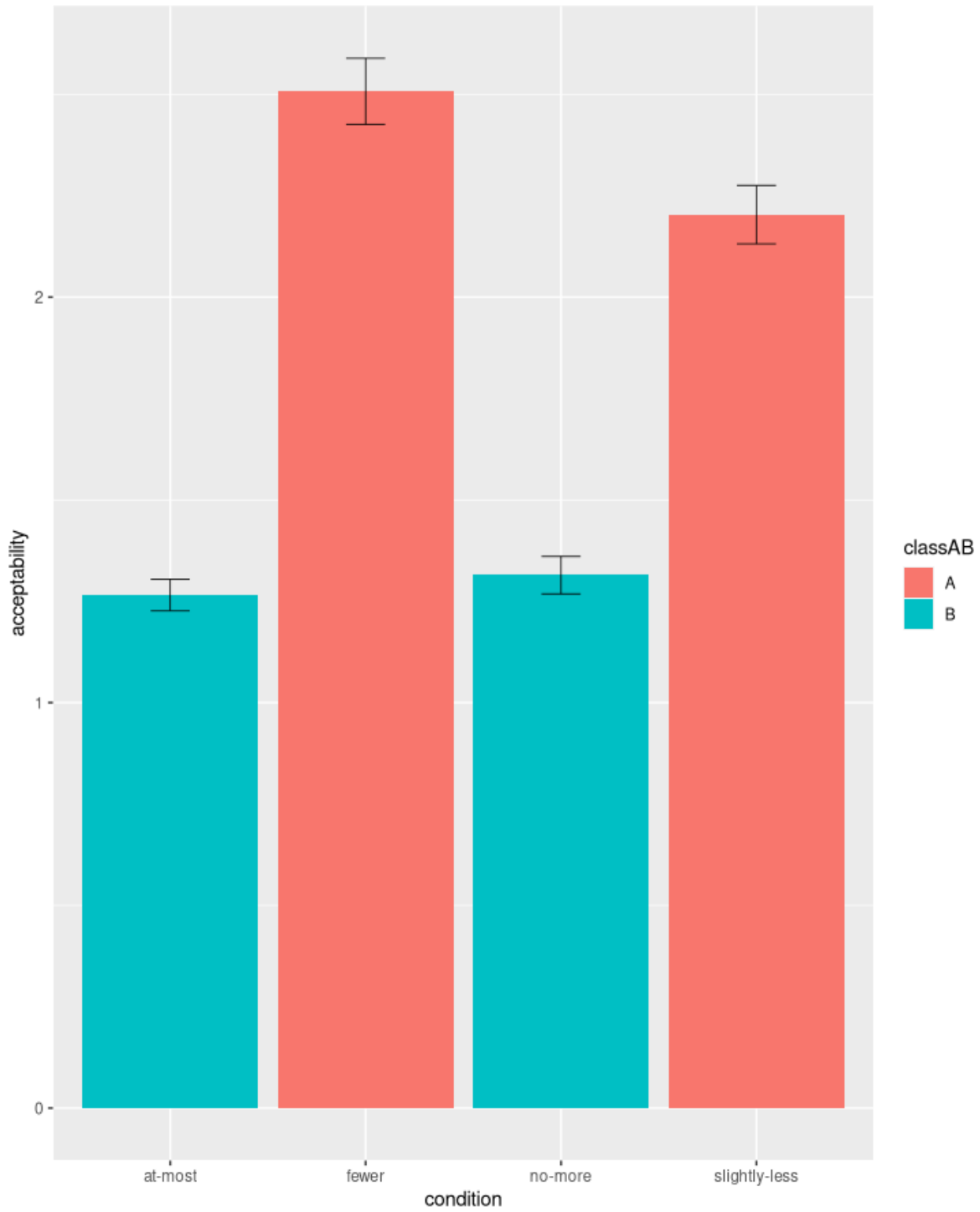


Figure 1: Boxplot of responses

4.2.5 Analysis

- the scope behaviour of Czech NMC is of a superlative-modifier profile
- following Nouwen 2008, we analyze German/Dutch *nicht mehr/niet meer* as a negative differential expressing that there is no positive difference in degree between the arguments of the comparative *more*:

$$[[\text{nicht mehr}]]\alpha = \lambda P. \neg \exists d' [max_d(P(d)) = \alpha + d']$$

- since the negative differential analysis is equivalent to the superlative-modifier at-issue semantics of *at most*:

$$\lambda P. max_d(P(d)) \leq \alpha$$

- applied to Czech experimental data, this correctly derives the similar scope behaviour of NMC and superlative modifiers
- the wide scope of the NMC/superlative modifiers then is:

$$max_d(\Diamond \text{contain}(\text{ChocBar}, d)) \leq 65g$$

- incompatible with Alex’s continuation and predicts low acceptability of NO-MORE and AT-MOST in the experiment
- weak surface scope ($\Diamond [max_d(\text{contain}(\text{ChocBar}, d)) \leq 65g]$), which allows a “more than” reading, is available only for comparative modifiers and explains the higher acceptability of FEWER and SLIGHTLY-LESS (compare Blok 2019)
- the scope behaviour of Czech NMC shows that semantically NMC behaves as superlative modifiers, despite its comparative morphology
- the experiment also supports the comparative-modifier vs. superlative-modifier theory presented by Kennedy 2015, where the distinction boils down to the type of ordering relation (strict vs. non-strict)
- NMC can, at least in languages like Czech, be interpreted as $\neg$ strict, resulting in ordering entailments of non-strict ordering. Regular differential quantifiers (SLIGHTLY-LESS) remain strictly ordered, thus comparative modifiers
- finally, we found three cross-linguistic types of NMC languages:

1. NMC as comparative modifiers, English type of NMC (bounding inferences and both scopes with respect to existential modals)
 2. NMC as superlative modifiers, Czech type of NMC (only $max_d > \Diamond$, lack of bounding inferences; compare Docekal 2017)
 3. languages where NMC, depending on its realization, behaves as comparative modifiers or as superlative modifiers (Hungarian according to Balazs Suranyi, p.c.)
- the variation is related to morpho-syntactic status: a constituent negation in NMC (Czech) behaves as a superlative modifier; a negative quantifier (English) in NMC leads to comparative modifiers

- our experiment shows that treating all NMCs uniformly as comparative modifiers is cross-linguistically untenable and that the distinction between comparative modifiers and superlative modifiers is not purely morphological
- Czech NMC contains both a comparative marker and a comparative standard marker, but unlike regular comparative-modifier differentials, Czech NMC acts as a superlative numeral modifier

Context: Alex is reading the following sentence on a chocolate bar packaging:

(2) Toto balení muze obsahovat:

- a. mene nez 60 gramu cukru
- b. nanejvys 60 gramu cukru
- c. ne vic nez 60 gramu cukru
- d. trochu mene nez 60 gramu cukru

‘This packaging can contain {a. fewer than / b. at most / c. no more than / d. slightly less than} 60 grams of sugar.’

Alex says: ‘So, in this chocolate bar there can be sometimes even 65 grams of sugar.’

5 Reports

```
library(plyr)
library(dplyr)

items <- read.csv2("data/clean_items.csv", encoding = "UTF-8", header = TRUE)

items <- items %>%
  mutate(condition = replace(condition, condition == "item-méně_než", "fewer")) %>%
  mutate(condition = replace(condition, condition == "item-nanejvýš", "at-most")) %>%
  mutate(condition = replace(condition, condition == "item-ne_víc_než", "no-more")) %>%
  mutate(condition = replace(condition, condition == "item-trochu_méně", "slightly-less")) %>%
  as.data.frame()

ddply(items, .(condition), summarise, Means = mean(rating1, na.rm = TRUE))
```

	condition	Means
1	at-most	1.262755
2	fewer	2.512755

```
3      no-more 1.311224
4 slightly-less 2.211735
```

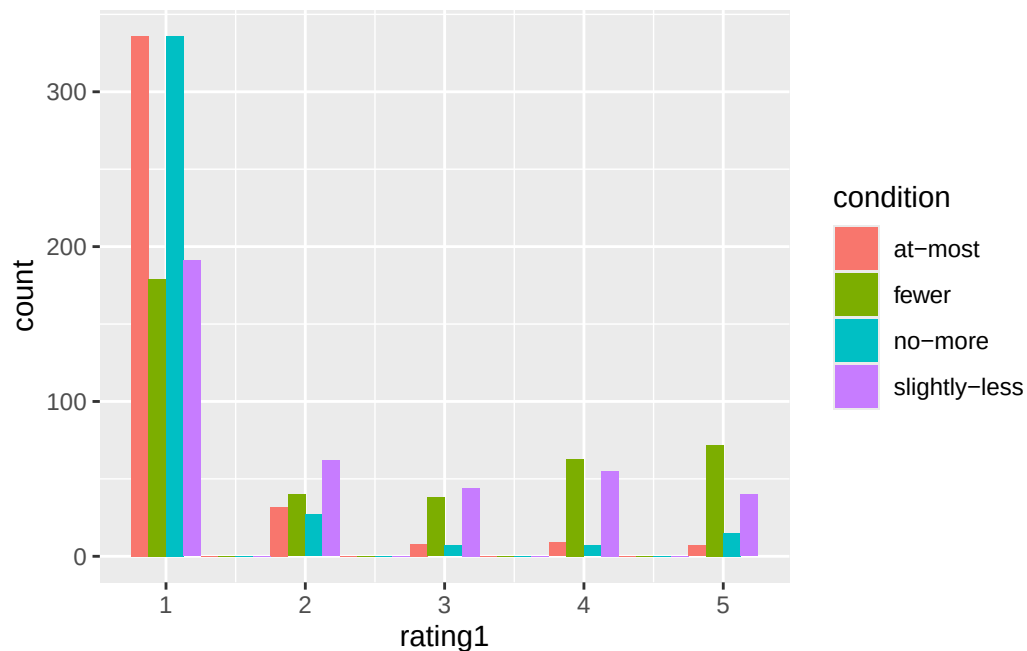
```
ddply(items, .(condition), summarise, Medians = median(rating1, na.rm = TRUE))
```

	condition	Medians
1	at-most	1
2	fewer	2
3	no-more	1
4	slightly-less	2

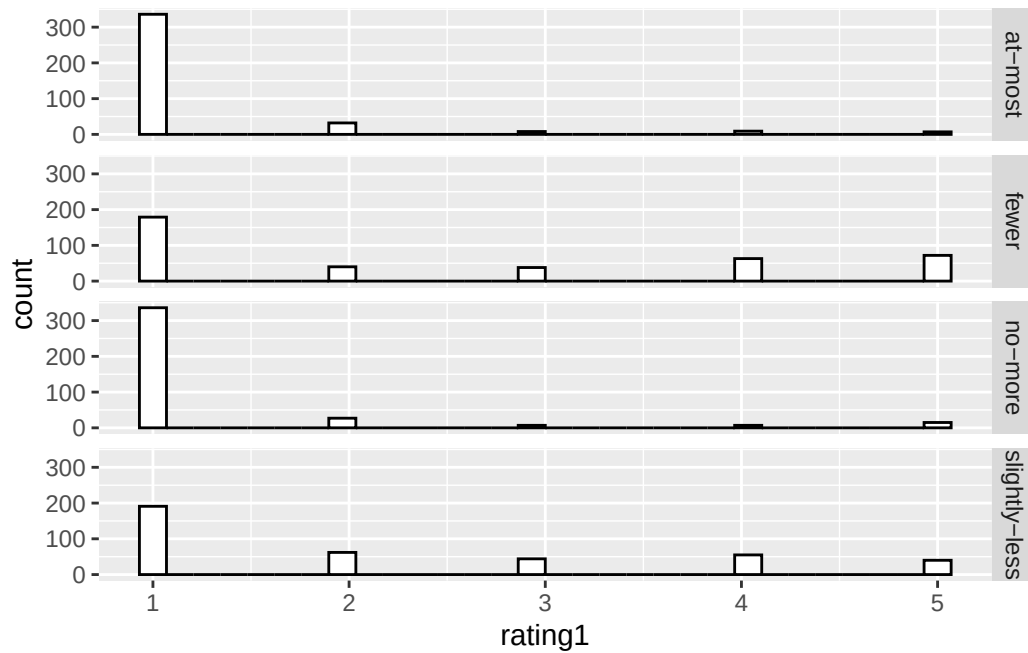
- graphs

```
library(ggplot2)
```

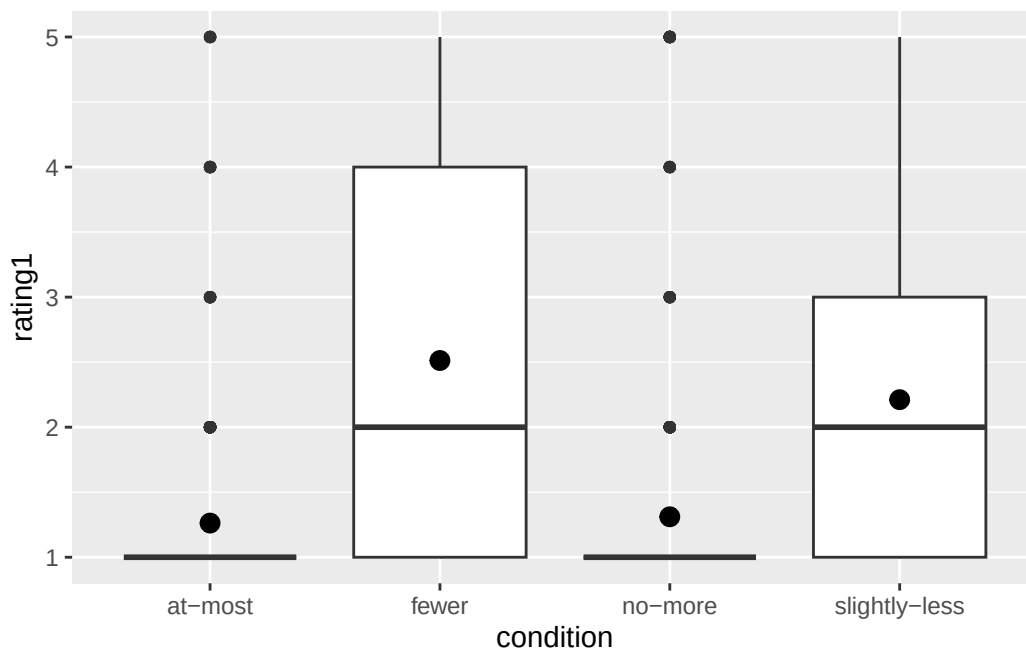
```
ggplot(items, aes(x = rating1, fill = condition)) +
  geom_histogram(position = "dodge", alpha = 1, binwidth = 0.5)
```



```
ggplot(items, aes(x = rating1)) +
  geom_histogram(fill = "white", colour = "black") +
  facet_grid(condition ~ .)
```



```
data.to.plot <- items
graph_to_plot <- ggplot(data.to.plot, aes(condition, rating1))
graph_to_plot + geom_boxplot() + stat_summary(fun.y = mean, geom = "point", size = 3)
```



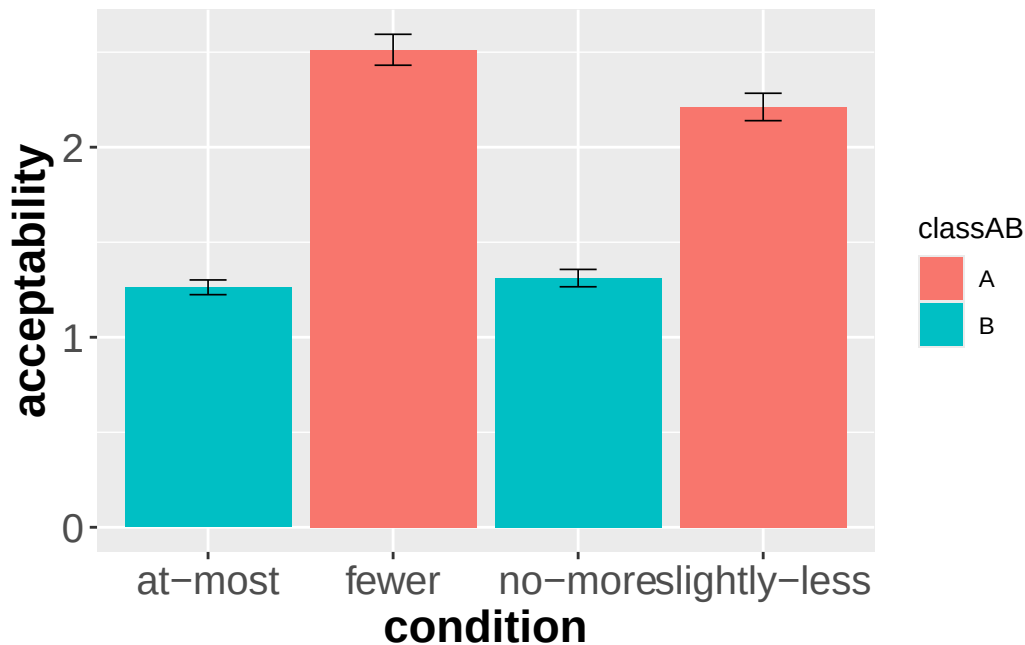
```

items$classAB <- "A"
items$classAB[items$condition == "at-most"] <- "B"
items$classAB[items$condition == "no-more"] <- "B"

p <- ggplot(items, aes(condition, rating1, fill = classAB)) +
  stat_summary(geom = "bar", fun.y = mean, position = "dodge") +
  stat_summary(geom = "errorbar", fun.data = mean_se, size = .3,
    width = .2,
    position = position_dodge(.9))

p + labs(y = "acceptability") +
  theme(axis.text = element_text(size = 15),
    axis.title = element_text(size = 17, face = "bold"))

```



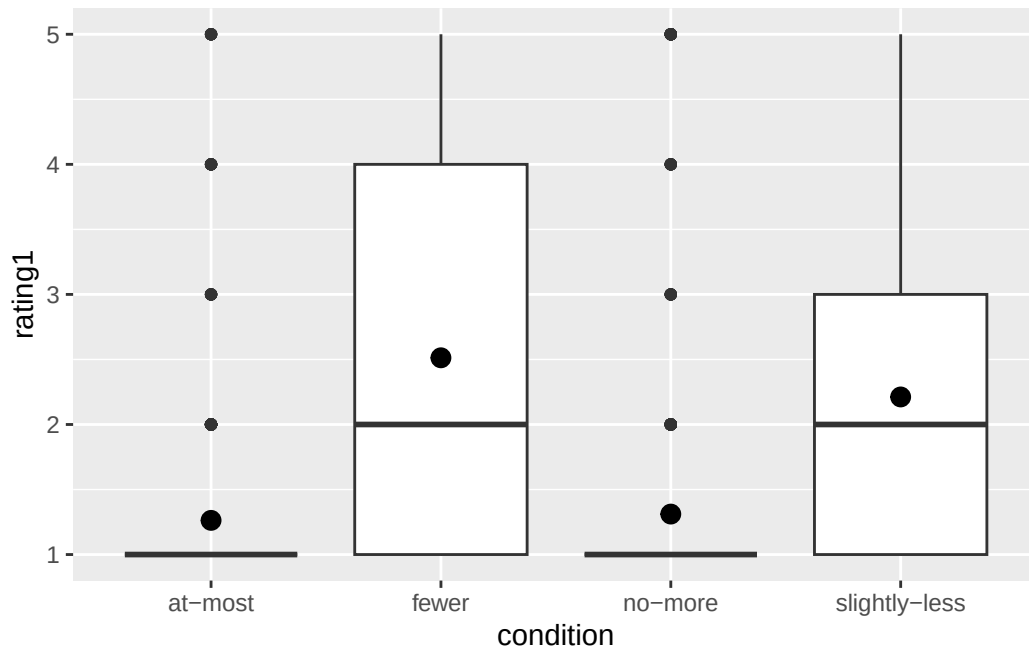
6 Models

```

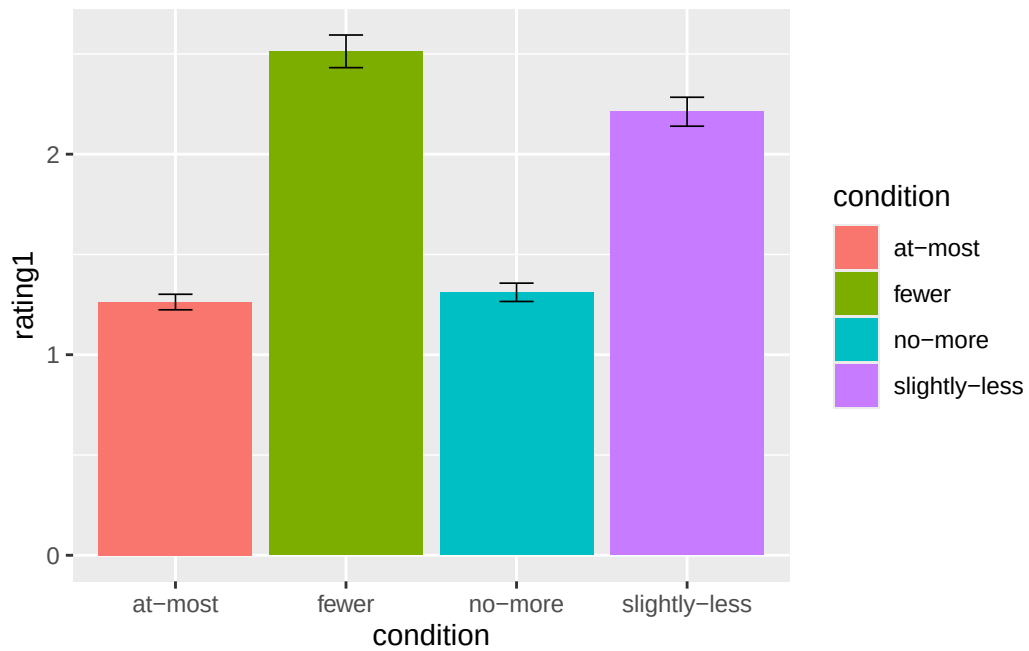
library(ggplot2)
data.to.plot <- items

graph_to_plot <- ggplot(data.to.plot, aes(condition, rating1))
graph_to_plot + geom_boxplot() + stat_summary(fun.y = mean, geom = "point", size = 3)

```



```
ggplot(items, aes(condition, rating1, fill = condition)) +
  stat_summary(geom = "bar", fun.y = mean, position = "dodge") +
  stat_summary(geom = "errorbar", fun.data = mean_se, size = .3,
    width = .2,
    position = position_dodge(.9))
```



```
library(dplyr)
```

```
N <- length(items$rating1)
N
```

```
[1] 1568
```

```
items.summary <- items %>%
  group_by(condition) %>%
  summarise(
    sd = sd(rating1),
    se = sd / sqrt(N),
    acceptability = mean(rating1)
  )
```

```
items.summary
```

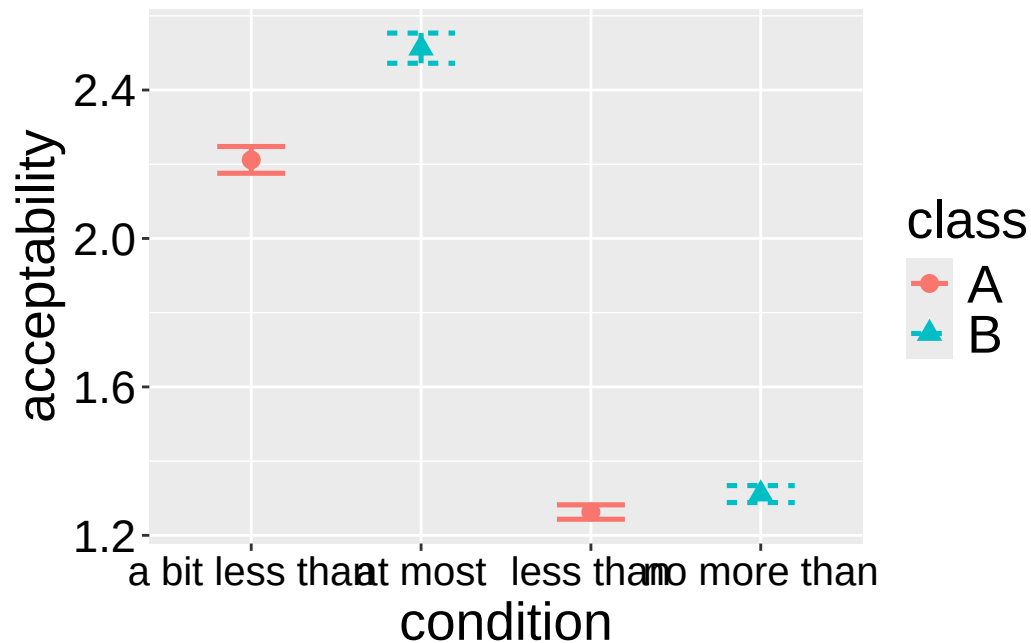
```
# A tibble: 4 x 4
  condition      sd      se acceptability
  <chr>         <dbl> <dbl>         <dbl>
1 at-most      0.767 0.0194         1.26
2 fewer        1.61  0.0407         2.51
3 no-more      0.905 0.0228         1.31
4 slightly-less 1.43  0.0361         2.21
```

```
items.summary[1, 1] <- "less than"
items.summary[2, 1] <- "at most"
items.summary[3, 1] <- "no more than"
items.summary[4, 1] <- "a bit less than"
```

```
items.summary$class <- NA
items.summary$class[1] <- "A"
items.summary$class[2] <- "B"
items.summary$class[3] <- "B"
items.summary$class[4] <- "A"
```

```
dodge <- position_dodge(width = 0.3)
qplot(condition, acceptability, colour = class, pch = class, lty = class,
  data = items.summary, geom = "point", size = I(3)) +
  geom_errorbar(aes(ymin = acceptability - se, ymax = acceptability + se),
    width = 0.4, size = I(0.9)) +
```

```
theme(axis.text.x = element_text(size = 15, colour = "black"),
      axis.text.y = element_text(angle = 0, hjust = 0.5, size = 17, colour = "black"),
      axis.title.y = element_text(size = 20),
      title = element_text(size = 20),
      legend.text = element_text(size = 20))
```



```
library(lmerTest)

items$condition <- as.factor(items$condition)
items$condition <- relevel(items$condition, ref = "at-most")

m1 <- lmer(as.numeric(rating1) ~ condition + (1 | participant) + (1 | item), data = items)

summary(m1)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: as.numeric(rating1) ~ condition + (1 | participant) + (1 | item)
Data: items

REML criterion at convergence: 4936.9

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.1683	-0.6481	-0.2050	0.4471	3.6096

Random effects:

Groups	Name	Variance	Std.Dev.
participant	(Intercept)	0.1414	0.3761
item	(Intercept)	0.1375	0.3708
Residual		1.2402	1.1137

Number of obs: 1568, groups: participant, 98; item, 16

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	1.271e+00	1.149e-01	2.942e+01	11.063	5.3e-12 ***
conditionfewer	1.242e+00	7.959e-02	1.452e+03	15.601	< 2e-16 ***
conditionno-more	3.996e-02	7.959e-02	1.452e+03	0.502	0.616
conditionslightly-less	9.322e-01	7.961e-02	1.452e+03	11.710	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	cndtnf	cndtnn-
conditinfwr	-0.346		
conditnn-mr	-0.346	0.500	
cndtnslght-	-0.346	0.500	0.500

```
library(emmeans)
```

```
emmeans(m1, list(pairwise ~ condition), adjust = "tukey")
```

\$`emmeans of condition`

condition	emmean	SE	df	lower.CL	upper.CL
at-most	1.27	0.115	29.4	1.04	1.51
fewer	2.51	0.115	29.4	2.28	2.75
no-more	1.31	0.115	29.4	1.08	1.55
slightly-less	2.20	0.115	29.4	1.97	2.44

Degrees-of-freedom method: satterthwaite

Results are given on the as.numeric (not the response) scale.

Confidence level used: 0.95

\$`pairwise differences of condition`

1	estimate	SE	df	t.ratio	p.value
---	----------	----	----	---------	---------

(at-most) - fewer	-1.242	0.0796	1452	-15.601	<0.0001
(at-most) - (no-more)	-0.040	0.0796	1452	-0.502	0.9586
(at-most) - (slightly-less)	-0.932	0.0796	1452	-11.710	<0.0001
fewer - (no-more)	1.202	0.0796	1452	15.095	<0.0001
fewer - (slightly-less)	0.309	0.0796	1452	3.888	0.0006
(no-more) - (slightly-less)	-0.892	0.0796	1452	-11.211	<0.0001

Note: contrasts are still on the as.numeric scale. Consider using
 regrid() if you want contrasts of back-transformed estimates.
 Degrees-of-freedom method: satterthwaite
 P value adjustment: tukey method for comparing a family of 4 estimates

```
items$classAB <- "A"
items$classAB[items$condition == "at-most"] <- "B"
items$classAB <- as.factor(items$classAB)

m2 <- lmer(as.numeric(rating1) ~ classAB + (1 | participant) + (1 | item), data = items)

summary(m2)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
 lmerModLmerTest]
 Formula: as.numeric(rating1) ~ classAB + (1 | participant) + (1 | item)
 Data: items

REML criterion at convergence: 5157.5

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.8526	-0.6835	-0.3082	0.4417	3.3690

Random effects:

Groups	Name	Variance	Std.Dev.
participant	(Intercept)	0.1285	0.3584
item	(Intercept)	0.1366	0.3696
Residual		1.4480	1.2033

Number of obs: 1568, groups: participant, 98; item, 16

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	2.00910	0.10526	20.39488	19.09	1.76e-14 ***
classABB	-0.73794	0.07022	1454.35268	-10.51	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

(Intr)

classABB -0.167

```
anova(m1, m2)
```

Data: items

Models:

m2: as.numeric(rating1) ~ classAB + (1 | participant) + (1 | item)

m1: as.numeric(rating1) ~ condition + (1 | participant) + (1 | item)

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
m2	5	5161.3	5188.1	-2575.7	5151.3			
m1	7	4937.8	4975.3	-2461.9	4923.8	227.5	2	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
items$classAB[items$condition == "no-more"] <- "B"
```

```
m3 <- lmer(as.numeric(rating1) ~ classAB + (1 | participant) + (1 | item), data = items)
```

```
summary(m3)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [

lmerModLmerTest]

Formula: as.numeric(rating1) ~ classAB + (1 | participant) + (1 | item)

Data: items

REML criterion at convergence: 4945.8

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-2.0147	-0.6657	-0.2035	0.4376	3.5794

Random effects:

Groups	Name	Variance	Std.Dev.
participant	(Intercept)	0.1407	0.3751
item	(Intercept)	0.1367	0.3698
Residual		1.2517	1.1188

Number of obs: 1568, groups: participant, 98; item, 16

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	2.35808	0.10761	22.83181	21.91	<2e-16 ***
classABB	-1.06693	0.05655	1454.41148	-18.87	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

(Intr)
classABB -0.263

```
anova(m1, m2, m3)
```

Data: items

Models:

```
m2: as.numeric(rating1) ~ classAB + (1 | participant) + (1 | item)
m3: as.numeric(rating1) ~ classAB + (1 | participant) + (1 | item)
m1: as.numeric(rating1) ~ condition + (1 | participant) + (1 | item)
```

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
m2	5	5161.3	5188.1	-2575.7	5151.3			
m3	5	4949.1	4975.9	-2469.6	4939.1	212.180	0	
m1	7	4937.8	4975.3	-2461.9	4923.8	15.316	2	0.0004723 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
items$Diff <- "NoDiff"
items$Diff[items$condition == "no-more" | items$condition == "slightly-less"] <- "Diff"
items$Diff <- as.factor(items$Diff)

m4 <- lmer(as.numeric(rating1) ~ classAB * Diff + (1 | participant) + (1 | item), data = items)

summary(m4)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [

lmerModLmerTest]

Formula: as.numeric(rating1) ~ classAB * Diff + (1 | participant) + (1 | item)

Data: items

REML criterion at convergence: 4936.9

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.1683	-0.6481	-0.2050	0.4471	3.6096

Random effects:

Groups	Name	Variance	Std.Dev.
participant	(Intercept)	0.1414	0.3761
item	(Intercept)	0.1375	0.3708
Residual		1.2402	1.1137

Number of obs: 1568, groups: participant, 98; item, 16

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	2.20337	0.11491	29.42257	19.175	< 2e-16 ***
classABB	-0.89223	0.07959	1452.25414	-11.211	< 2e-16 ***
DiffNoDiff	0.30942	0.07959	1452.25414	3.888	0.000106 ***
classABB:DiffNoDiff	-0.34938	0.11252	1452.10172	-3.105	0.001939 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	clsABB	DffNDf
classABB	-0.346		
DiffNoDiff	-0.346	0.500	
clssABB:DND	0.245	-0.707	-0.707

```
anova(m1, m2, m3, m4)
```

Data: items

Models:

```
m2: as.numeric(rating1) ~ classAB + (1 | participant) + (1 | item)
m3: as.numeric(rating1) ~ classAB + (1 | participant) + (1 | item)
m1: as.numeric(rating1) ~ condition + (1 | participant) + (1 | item)
m4: as.numeric(rating1) ~ classAB * Diff + (1 | participant) + (1 | item)
```

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
m2	5	5161.3	5188.1	-2575.7	5151.3			
m3	5	4949.1	4975.9	-2469.6	4939.1	212.180	0	
m1	7	4937.8	4975.3	-2461.9	4923.8	15.316	2	0.0004723 ***
m4	7	4937.8	4975.3	-2461.9	4923.8	0.000	0	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
items$classAB[items$condition == "no-more"] <- "A"

m5 <- lmer(as.numeric(rating1) ~ classAB * Diff + (1 | participant) + (1 | item), data = items)

summary(m5)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]
Formula: as.numeric(rating1) ~ classAB * Diff + (1 | participant) + (1 | item)
Data: items

REML criterion at convergence: 5054.3

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.0655	-0.6653	-0.2473	0.4431	3.4743

Random effects:

Groups	Name	Variance	Std.Dev.
participant	(Intercept)	0.1348	0.3672
item	(Intercept)	0.1387	0.3724
Residual		1.3465	1.1604

Number of obs: 1568, groups: participant, 98; item, 16

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	1.75728	0.10845	22.67242	16.20	5.88e-14 ***
classABB	-1.24116	0.08292	1453.27185	-14.97	< 2e-16 ***
DiffNoDiff	0.75525	0.07183	1453.38009	10.52	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	clsABB
classABB	0.000	
DiffNoDiff	-0.221	-0.577

fit warnings:

fixed-effect model matrix is rank deficient so dropping 1 column / coefficient

```
summary(m4)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: as.numeric(rating1) ~ classAB * Diff + (1 | participant) + (1 |
      item)
Data: items
```

```
REML criterion at convergence: 4936.9
```

```
Scaled residuals:
```

	Min	1Q	Median	3Q	Max
	-2.1683	-0.6481	-0.2050	0.4471	3.6096

```
Random effects:
```

Groups	Name	Variance	Std.Dev.
participant	(Intercept)	0.1414	0.3761
item	(Intercept)	0.1375	0.3708
Residual		1.2402	1.1137

```
Number of obs: 1568, groups: participant, 98; item, 16
```

```
Fixed effects:
```

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	2.20337	0.11491	29.42257	19.175	< 2e-16 ***
classABB	-0.89223	0.07959	1452.25414	-11.211	< 2e-16 ***
DiffNoDiff	0.30942	0.07959	1452.25414	3.888	0.000106 ***
classABB:DiffNoDiff	-0.34938	0.11252	1452.10172	-3.105	0.001939 **

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Correlation of Fixed Effects:
```

	(Intr)	clsABB	DffNDf
classABB		-0.346	
DiffNoDiff	-0.346		0.500
classABB:DND	0.245	-0.707	-0.707

```
anova(m4, m5)
```

```
Data: items
```

```
Models:
```

```
m5: as.numeric(rating1) ~ classAB * Diff + (1 | participant) + (1 | item)
```

```

m4: as.numeric(rating1) ~ classAB * Diff + (1 | participant) + (1 | item)
      npar      AIC      BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
m5      6 5056.6 5088.8 -2522.3    5044.6
m4      7 4937.8 4975.3 -2461.9    4923.8 120.81  1 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```